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Development of AgriGrade AI -Deep Learning-Based Vegetable Quality & Grade Detection System

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Description

AgriGrade AI is a research-driven project developed to address the critical issue of subjectivity and inconsistency in manual agricultural grading. In many global supply chains, the lack of standardized quality tools leads to significant post-harvest losses and unfair pricing for small-scale farmers. This project demonstrates how high-accuracy Deep Learning models can be democratized through cloud-native, mobile-first web technologies, providing an objective "Standard of Truth" for produce quality. The Human Impact & Motivation During my research, I observed that manual grading is often a bottleneck for farmers. What one inspector sees as 'Grade A' produce might be downgraded by another at the market gate, directly affecting a farmer's livelihood. I built AgriGrade AI to put a professional-grade inspection tool directly into the pockets of those who need it most. By bridging the gap between high-end Convolutional Neural Networks (CNNs) and the real world, we can help farmers make data-driven decisions about their produce in real-time, reducing the power imbalance in the supply chain. Technical Methodology & Innovation The system employs a multi-layered architectural approach designed for edge-compatibility: Computer Vision Core: Utilizing Transfer Learning via the MobileNetV2 architecture, the model was trained on specialized agricultural datasets to achieve a 98.9% validation accuracy. It classifies produce into a three-tier hierarchy: Grade A (Prime), Grade B (Transition), and Grade C (Degraded). Phyto-Metric Feature Analysis: Unlike generic image classifiers, this AI specifically targets biological marke

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20CS295– Mini Project I

"Development of AgriGrade AI – Deep Learning-Based Vegetable Quality & Grade Detection System"

Type of Project: Application
Expected Project Outcome : Product

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